Noah R. B. Hansen

MECHANICAL ENGINEERING STUDENT

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PROFILE

Mechanical engineering student with a strong analytical mindset and experience with complex technical systems, reliability, structured problem-solving, technical documentation, data analysis, and troubleshooting. Comfortable working across hardware, software, and physical systems.

EDUCATION

BSc in Engineering (Mechanical Engineering)
Aarhus University
Aarhus, Denmark — *Aug. 2024 – Present*

- Member of the Education Committee.
- Provisional GPA: 10.5

HTX – Mat A/Fys A
Vejle Tekniske Gymnasium
Vejle, Denmark — *Aug. 2020 – Jun. 2023*

- Board member of the student council.
- Highest average grade in the nation; weighted GPA: 12.6.

SELECTED ACHIEVEMENTS

- Highest weighted GPA in the nation (2023)
- Danish Physics Olympiad national final qualification (2022)
- Georg Mohr competition winner seminar attendee (2022)
- Honourable mention, Projekt Forskerspirer (2022)

SKILLS

Technical	Electromechanical troubleshooting, DC motors, gearboxes, drivetrains, pumps, sensors, maintenance
Materials & testing	Four-point probe measurements, cryogenic handling, experimental data analysis
Tools	AutoCAD, Solidworks, EES, MDSolids, Mathcad, LaTeX, MS Office, Inkscape
Programming	Python, MATLAB, Lua, scripting
Languages	Danish (native), English (C2), German (B1)

WORK EXPERIENCE

Maintenance Assistant, LEGOLAND — *Dec. 2023 – Dec. 2025*
Billund, Denmark

- Performed independent electromechanical troubleshooting and maintenance on 24 V mobility-scooter-based electric vehicles used in LEGOLAND's traffic school.
- Completed full overhaul/restoration work on approximately 18 traffic school cars, including DC motor refurbishment, brush replacement, gearbox servicing, brake/speeder adjustment, bushings, wheels, and fiberglass bodywork.
- Maintained Miniland animations involving 12 V DC motors, gearboxes, charging wires, PCBs, sensors, batteries, drivetrain tolerances, belts, chain drives, pumps, and fuse boxes.
- Diagnosed recurring failure on a controllable Femern Belt crane animation and developed a reset procedure; proposed a welded spool-guide modification that substantially reduced repeat faults on the crane and sister crane.
- Supported daily operation of Miniland by monitoring faults, restarting systems, and ensuring mechanical/electrical attractions remained operational.

Facility Assistant, LEGOLAND — *Jun. – Dec. 2022 & Aug. – Dec. 2023*
Billund, Denmark

- Inspected approximately 193 technical rooms and infrastructure areas using digital checklists, creating work orders directly when issues could not be resolved on site.
- Supported safety and fire-compliance improvements by identifying deviations, following up on work orders, and helping ensure technical rooms met updated standards.
- Installed and dismantled approximately 400 outdoor wash lights for seasonal Christmas lighting, with attention to weatherproof cable routing and prevention of water ingress.

Tutor, Syddansk Erhvervsskole — *Aug. 2021 – Jun. 2023*
Vejle, Denmark

- Provided peer tutoring as a part of the schools homework cafe.

PROJECTS & RESEARCH

YBCO superconductors project
Investigated high-temperature superconductivity through Meissner-effect and flux-pinning demonstrations, four-point probe measurements across the critical temperature, and Python-based plotting, curve fitting, and numerical analysis. Received grade 12.

Projekt Forskerspirer
Developed an experimental research proposal on tuning the plasmon resonance frequency of semiconducting nanoparticles through charge-carrier-density control, involving doping, Raman scattering, and spectrophotometry. Received an honourable mention.

Tungsten — author and maintainer
Developed an open-source Neovim plugin that parses LaTeX mathematics and evaluates expressions directly in LaTeX documents using computational backends. Demonstrates systems thinking, technical documentation, release management, and complex problem decomposition.

DødsBo — tractor-pulling vehicle
Co-owned and rebuilt a garden-pulling tractor for a 350 kg competition class, including AutoCAD-based part design, metal fabrication, anti-tip brackets, weight-holder assemblies, drivetrain modification, and practical testing.